Multi-Layer Swarm Quadrotor Architecture for Autonomous Non-Convex Geodesic Mapping with Fault-Tolerant Cooperative Coverage

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Abstract, Autonomous swarm quadrotor systems have emerged as a promising solution for large-scale geodetic mapping in complex environments. This paper proposes a three-layer control architecture for resilient non-convex geodesic mapping using a quadrotor swarm. The high-level layer employs Voronoi-based non-convex region partitioning integrated with Boustrophedon path planning and Fault-Tolerant Cooperative Coverage (FT-CC) for adaptive task redistribution under failures. The mid-level layer utilizes Optimal Reciprocal Collision Avoidance (ORCA) for decentralized collision-free navigation. The low-level layer employs a cascaded PID control architecture, comparing optimal LQR synthesis with robust H_∞ synthesis (HARE) for attitude and position tracking under external disturbances. The framework is validated through high-fidelity simulation in ROS2 Lyrical, Gazebo Harmonic, and PX4 SITL. Expected results indicate 90–95% coverage of non-convex regions with zero inter-robot collisions, demonstrating the potential of the proposed architecture for real-world geodetic applications.

1. INTRODUCTION

Swarm quadrotor systems have gained significant attention for autonomous geodetic mapping, with applications including disaster response mapping of post-earthquake or landslide areas, precision agriculture through multispectral crop monitoring, environmental surveillance for forest fire and pollution detection, and large-scale infrastructure inspection of pipelines and bridges. In these scenarios, quadrotors equipped with optical cameras, multispectral sensors, or radar altimeters can effectively partition a given territory among multiple agents, enabling rapid data acquisition over large areas through coordinated coverage. These applications demand efficient region partitioning, robust flight control under uncertain environmental conditions, and resilient operation despite individual vehicle failures.

Despite progress in each domain individually, several critical gaps remain. First, most coverage algorithms assume convex operational regions, whereas real-world mapping areas are inherently non-convex with obstacles and irregular boundaries. Second, end-to-end pipelines from region partitioning to low-level rotor control are scarce, causing integration mismatches between planning and execution layers. Third, the systematic integration of low-level robust control, mid-level collision avoidance, and high-level coverage planning within a unified framework has not been adequately addressed. Fourth, fault-tolerant mechanisms in cooperative coverage for quadrotor swarms remain underexplored, particularly under communication or actuator failures. Fifth, validation in realistic simulation environments with full quadrotor dynamics is still limited, with most studies relying on simplified kinematic models. To address these gaps, this paper proposes a three-layer architecture integrating Voronoi-based non-convex partitioning, Boustrophedon path planning, FT-CC, ORCA-based collision avoidance, and cascaded PID-LQR / PID-H_∞ robust control, validated through high-fidelity ROS2-Gazebo-PX4 simulation.

2. METHODS (OR APPROACH)

The proposed architecture comprises three hierarchical layers operating in a closed loop, as illustrated in Figure 1. The high-level layer receives the non-convex region boundary and initial drone positions as inputs. Voronoi tessellation adapts to non-convex geometries through obstacle-aware distance metrics, partitioning the region into sub-regions assigned to individual drones. Boustrophedon sweeping generates optimal coverage paths for each sub-region, minimizing overlap and energy consumption. The FT-CC module continuously monitors drone health status and redistributes coverage responsibilities upon detection of actuator or communication failure, ensuring mission-level resilience. Generated waypoints are passed to the mid-level layer.

The mid-level layer implements ORCA, a decentralized collision avoidance protocol. Each drone broadcasts its position and velocity to neighbors, computes velocity obstacles relative to other agents, and selects a reciprocal collision-free velocity that preserves progress toward its assigned waypoint. The resulting velocity commands are forwarded to the low-level layer.

The low-level layer executes the received velocity commands through a cascaded PID control architecture. Both PID-LQR and PID-H_∞ share the identical cascaded topology, consisting of an outer position control loop (x, y, z) generating reference attitude angles and an inner attitude control loop (ϕ, θ, $\dot{\psi}$) regulating quadrotor torques. The distinction lies in gain synthesis: PID-LQR solves the standard continuous Algebraic Riccati Equation (ARE), whereas PID-H_∞ solves the H_∞ Riccati Equation (HARE) incorporating an explicit H_∞ disturbance attenuation bound γ. Control outputs are converted to motor PWM signals and applied to the quadrotor plant simulated in the Gazebo Harmonic physics engine running PX4 SITL firmware. State feedback (pose, velocity, IMU measurements) propagates upward through all layers, enabling ORCA to update collision avoidance decisions and FT-CC to detect faults and trigger coverage redistribution.

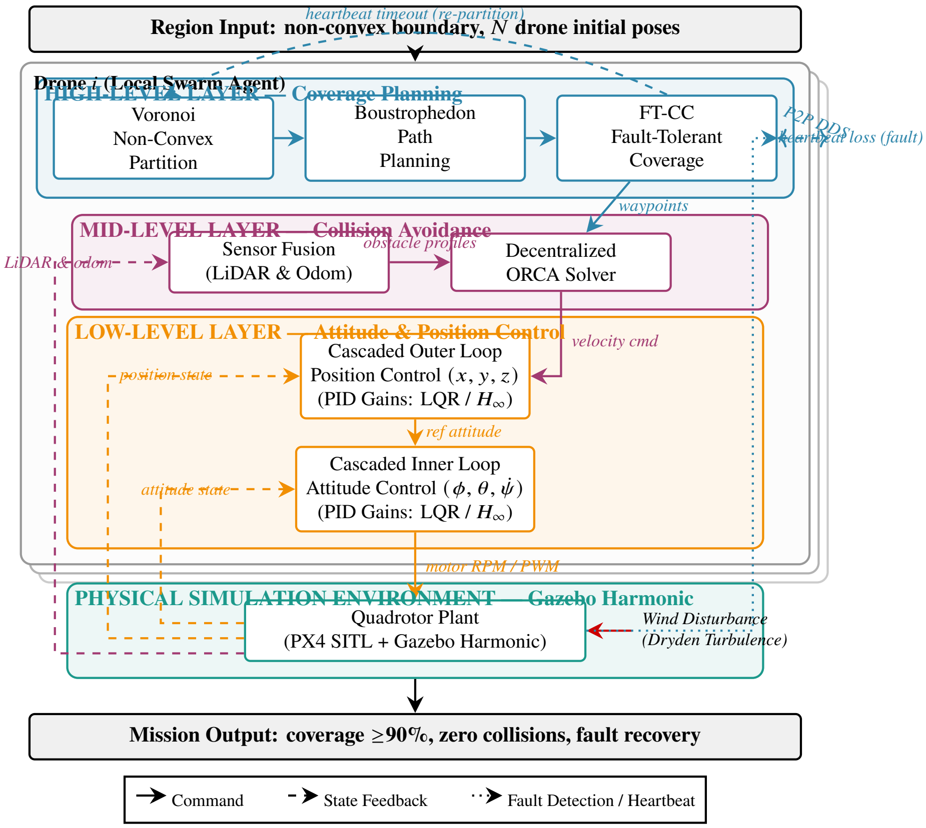

Figure 1 Proposed three-layer control architecture for resilient autonomous swarm quadrotor geodesic mapping.

3. PRELIMINARY RESULTS

The proposed framework is currently under evaluation through high-fidelity simulation using ROS2 Lyrical, Gazebo Harmonic, and PX4 SITL across multiple mission scenarios involving non-convex regions with obstacles. Expected performance includes 90–95% coverage completion of non-convex regions, zero inter-robot collisions throughout the mission duration, and successful fault recovery maintaining at least 85% coverage after simulated actuator or communication failure of one or more drones. The PID-Hinf position controller is expected to outperform PID-LQR in trajectory tracking accuracy under wind disturbance, with lower root-mean-square error (RMSE) in hover and waypoint navigation. Quantitative metrics including coverage ratio, collision count, trajectory tracking RMSE, and fault recovery time will be reported in the full paper.

4. CONCLUSION

This paper presents a three-layer control architecture for autonomous non-convex geodesic mapping by resilient quadrotor swarms. The main contributions include: (i) an end-to-end pipeline integrating non-convex Voronoi partitioning with Boustrophedon path planning and FT-CC fault tolerance at the high level; (ii) decentralized ORCA-based collision avoidance at the mid level; (iii) PID-LQR and PID-Hinf robust cascade control at the low level; and (iv) validation through a high-fidelity ROS2-Gazebo-PX4 simulation framework. The architecture addresses critical gaps in non-convex coverage, multi-layer integration, fault tolerance, and realistic simulation validation. Future work will focus on real-world outdoor flight experiments and extension to heterogeneous multi-robot teams with diverse sensor payloads.

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